Analysis of the Muon Lifetime and Nuclear Capture Rates in Carbon and Oxygen-dominated Environments

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Abstract

Detection of muon interactions and decays in a hydrocarbon scintillator setup equipped with photomultiplier tubes enables us to experimentally measure the lifetime of muons. By employing time-window selection, filtering out coincidences in signals, and accounting for negative muon capture rates, we obtain a muon lifetime of $\tau_\mu = (2.221 \pm 0.237) \times 10^{-6}$ s well within one standard deviation from the accepted result $\tau_{\text{true}} = 2.197 \times 10^{-6}$ s. Further confirmation of the lifetime using muon decays in a water tank also provides insight into the nuclear capture rate in this oxygen-dominated environment $\lambda^O_{\text{cap}} \simeq (0.9 \pm 0.2) \times 10^5\,\text{s}^{-1}$. The simultaneous treatment of both datasets—oxygen and carbon media—allows verification of the theoretical model (predicting $\lambda_{\text{cap}} \propto Z^3$ for light nuclei), obtaining $(\lambda^O_{\text{cap}}/\lambda^C_{\text{cap}}) \simeq 2.25 \sim (8/6)^3 = (Z_O/Z_C)^3$. In addition, we examine the possibility of determining the $\mu^+ : \mu^-$ ratio in the atmosphere from the same data. Finally, we address the issue of different detector efficiencies between the two experiments and suggest potential improvements.

Keywords: muon, decays, lifetime, nuclear capture rate, oxygen and carbon, detector

1 Introduction

The muon was discovered in 1936 by Carl D. Anderson and Seth Neddermeyer during their studies of cosmic rays using cloud chambers [1]. They observed a new charged particle behaving like a heavy electron — carrying the same electric charge and spin but having a much larger mass, yet lower than that of a proton. However, its existence was initially puzzling, as it did not participate in strong nuclear interactions. This was resolved by the lepton–hadron distinction and the concept of particle generations.

Within this revised framework of the Standard Model, the muon is now understood to be a second-generation lepton, governed by the weak and electromagnetic interactions but not by the strong force. It has a mass of 206.77 m_e and, unlike the electron, the muon is unstable, decaying via the weak interaction with a mean lifetime of 2.197×10^{-6} s into an electron and two neutrinos:

$$\mu^- \to e^- + \bar{\nu}_e + \nu_\mu \qquad (\mu^+ \to e^+ + \nu_e + \bar{\nu}_\mu)$$

At the Earth's surface, muons are produced primarily through the decay of pions and kaons generated by strong-interaction collisions of cosmic rays with nuclei in the upper atmosphere. These parent particles do not reach sea level, in spite of the associated time dilation, but muons do. This penetration of muons to sea level makes them readily observable in ground-based experiments, and was one of the earliest pieces of evidence for time dilation [2].

One other key aspect of muon interactions is that of nuclear capture. As these particles are slowed down in matter, the negative muons can drop into atomic shells, replacing orbital electrons. Due to their larger mass, their atomic orbitals lie much closer to the positive nucleus than those of electrons and can thus be captured by it, forming so-called *muonic atoms* [3]. Positively charged muons cannot form bound states around the nucleus, being repelled, and so do not undergo nuclear capture.

$$\mu^- + p \to n + \nu_\mu$$

The aim of this project is to examine these competing processes: muon decay and nuclear capture. We begin by describing the setup used for detections and the methods employed throughout the analysis. Then, we extract the muon lifetime from the decay data in two different media and compare it with the accepted values, making corrections as needed. Afterward, we explore the different capture rates of negative muons in the two environments and verify the theoretical dependence. We finally perform the relevant error analysis and draw conclusions valuable to this and future experiments.

2 Methods and Apparatus

2.1 Theoretical Model

Muons ionize nearby atoms, losing energy in the process and eventually coming to rest in matter, where they either decay via the charged-current weak interaction or, if negatively charged, undergo nuclear capture. The observed number of stopping muons will follow an exponential distribution:

$$N(t) = N_0 e^{-t/\tau_{\text{obs}}}, \tag{1}$$

where the total disappearance rate is given by the sum of the decay and capture rates

$$\frac{1}{\tau_{\text{obs}}} = \lambda_{\text{tot}} = \lambda_0 + f_- \lambda_{\text{cap}} \, . \tag{2}$$

Here, f_- accounts for the fraction of μ^-, hence the true lifetime of the muon will then be

$$\tau_\mu = \frac{1}{\lambda_0} \, . \tag{3}$$

To properly account for the two charge populations, it is more appropriate to use a bi-exponential model for our stopping muon distributions:

$$N(t) = N_0 \left[(1 - f_-) e^{-\lambda_0 t} + f_- e^{-(\lambda_0 + \lambda_{\text{cap}})t} \right] \, . \tag{4}$$

We use the first model given by Equation (1) in the initial calculation for a single homogeneous environment in order to estimate the lifetime and then correct for it. The second model in Equation (4) is used when treating a different nuclear medium, and then in the simultaneous treatment of both datasets to identify the corresponding capture rates.

These rates are expected to differ between media*, labeled by j, according to Fermi's Golden Rule:

$$\lambda^j_{\text{cap}} = \frac{2\pi}{\hbar} \left| \langle f | H_{\text{int}} | i \rangle^j \right|^2 \rho^j(E_f). \tag{5}$$

The matrix element factorizes into the muon wavefunction evaluated at the nucleus, and a nuclear matrix element which varies slowly with Z for light nuclei:

$$\langle f | H_{\text{int}} | i \rangle^j \propto G_f \, \psi^j_\mu(0) \, M^j_{\text{nucl}} \, . \tag{6}$$

The muon of mass m_μ bound to a nucleus of charge $+Z_j \, e$, has the ground state wavefunction

$$\psi^j_{1s}(r) = \frac{1}{\sqrt{\pi}} \left(\frac{Z_j}{a_\mu} \right)^{\frac{3}{2}} e^{-Z_j r / a_\mu} \, , \tag{7}$$

where $a_\mu = a_0 \frac{m_e}{m_\mu}$ and a_0 is the Bohr radius. Evaluated at $r = 0$, we get

*The superscript j is used to distinguish between media, as we expect different capture rates λ^j_{cap} and different interaction matrix elements $\langle f | H_{\text{int}} | i \rangle^j$ for each. The density of final states $\rho^j(E_f)$ also varies with the specific nucleus, yet that variation is slow compared to the dominant contribution from the interaction element.

$$|\psi^j_{1s}(0)|^2 = \frac{1}{\pi}\left(\frac{Z_j m_\mu}{a_0 m_e}\right)^3 \Rightarrow |\psi^j_{1s}(0)|^2 \propto Z_j^3\,. \tag{8}$$

And thus we obtain for the rate:

$$\lambda^j_{\text{cap}} \propto G_f^2 \underbrace{|\psi^j_{1s}(0)|^2}_{\propto Z_j^3} \underbrace{|M^j_{\text{nucl}}|^2}_{\text{slowly varying}}\,. \tag{9}$$

Consequently, we expect our capture rate measurements to obey a dependence like $\lambda^j_{\text{cap}} \propto Z_j^3$.

2.2 Experimental Setup

2.2.1 Scintillator Layout

For the first part of the experiment, a layout of three scintillators is used. These are made of hydrocarbon material and are equipped with photomultiplier tubes (PMTs), such that when a muon ionizes atoms in the scintillator, the emitted light falls on the photocathode of these PMTs, liberating electrons. The electrons are then accelerated in a strong electric field, successively hitting dynodes which emit more electrons. In the end, 10^6–10^7 electrons reach the PMT anode for each electron emitted at the cathode, and are then converted to electric signals through a high-precision oscilloscope (*picoscope*).

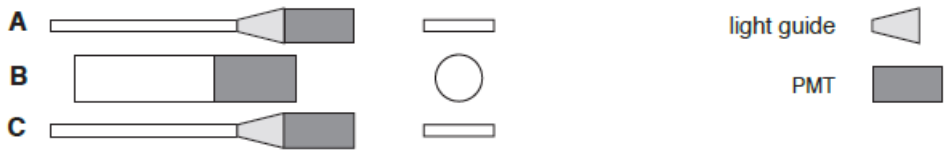


Figure 1: Schematic layout of the detectors A, B and C (taken from [4])

The detectors are placed one on top of the other, as illustrated in Figure 1, with the top and bottom ones being thin slabs (A & C), and the one in between being a thick cylinder (B). This setup is useful for recording muons, whose flux is about one per cm^2 per minute, and can differentiate between stopping (S) and through-going (T) muons. A and C are thin and do not stop many muons; however, B does, enabling us to check whether there is a correlated signal on the bottom detector, or whether the muon stopped along the way before reaching it — either through decay or muon capture.

2.2.2 Water Tank

The second setup used involves an alternative environment where muons can decay or be captured: a water tank (Figure 2). A similar slab detector D is placed on top of the container. The tank comes with a detector of its own, which registers any Cherenkov radiation emitted by particles traveling faster than the phase velocity of light in that medium.

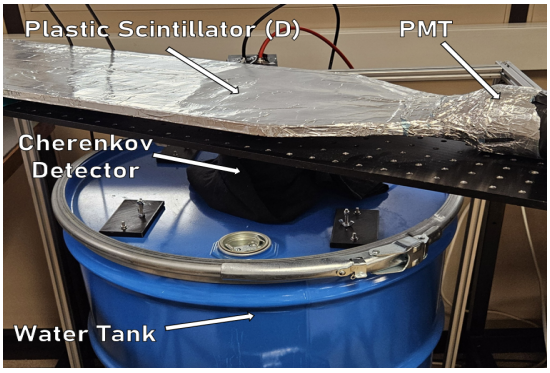


Figure 2: Layout of the water tank setup

While the mechanism is explained in detail in Ref. [5], it is crucial that both muons and their decay products (electrons or positrons) emit Cherenkov radiation as they go through the tank, all being

highly energetic. Having no possibility of placing another detector underneath the tank, we utilize this principle to set physical time windows between successive signals so we can discriminate between S and T muons.

2.3 Data Acquisition and Event Selection

Once the setup is in place and all detectors connected to their corresponding picoscope channels, we set the trigger on the top detector in both cases and impose a threshold on signal amplitude of 100 mV to retain only meaningful signals. We begin the data acquisition and let it run for one day for the first setup, and one week for the second. The interactions are visualized and plotted live in the *Picoscope 6* software [6], and all of the signals are saved with their timestamps and intensities in a file generated by the oscilloscope program.

The detector data is then filtered in order to highlight the actual muon interactions by employing a series of cuts. These cuts will generally remain the same throughout the analysis, with minor modifications as follows:

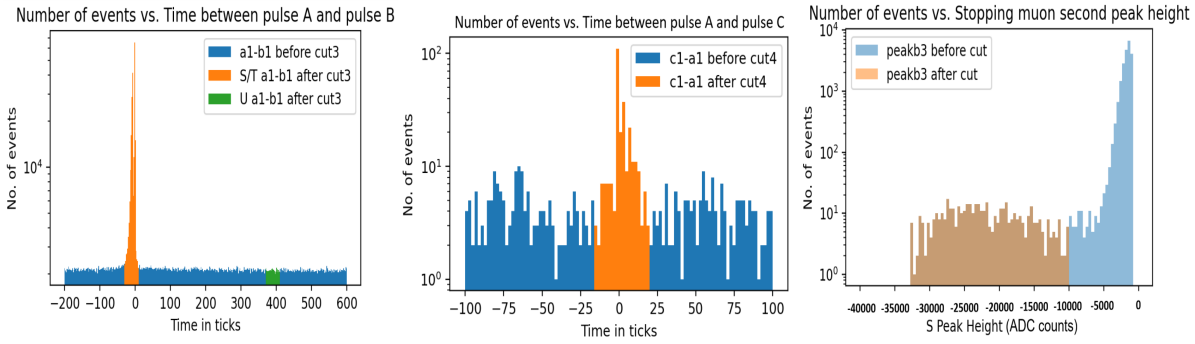
Cuts 1 and 2 simply check whether relevant pulses exist on the first and second detector (Note: this implies that while the names of the detectors and channels are the same in the first setup, channels A and B will correspond to detector D and the Cherenkov detector, respectively, in the water tank case).

Cut 3 is employed so we can select an effective signal window, highlighted in orange in Figure 3a, and an uncorrelated (U) signal background farther away, highlighted in green, to subtract.

Cut 4 is similar, eliminating coincidences in C, by selecting a window for T muons, and a larger one for S and U which do not hit detector C. This time interval is considered in ticks of 8 ns and taken between A and C, as can be seen in Figure 3b.

Cut 5 checks for another pulse on the B channel, since it is only the S muons that should produce this from decays.

Cut 6 imposes a threshold on this second B pulse in order to leave only the true muon decays and filter out any accidental counts or after-pulses, highlighted in Figure 3c.



(a) Cut 3 profile **(b)** Cut 4 profile **(c)** Cut 6 profile

Figure 3: Sample profiles for the filters on pulse timing and height

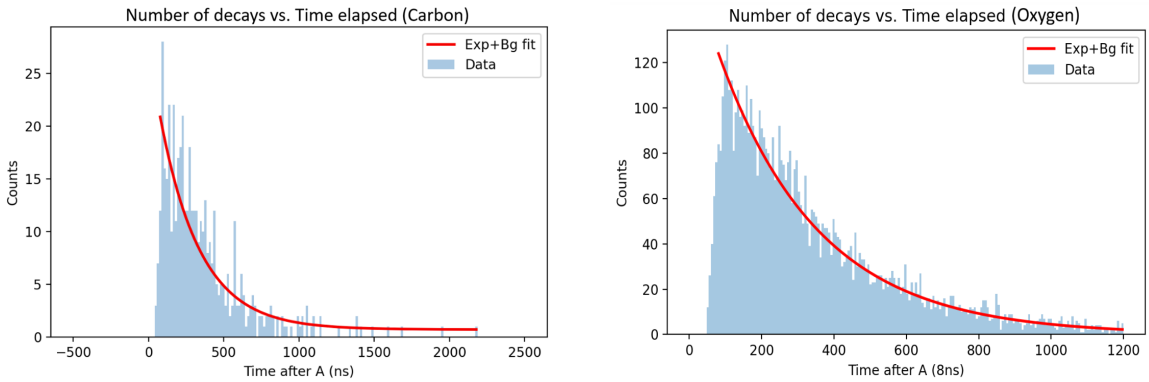
3 Results and Analysis

Having collected approximately (3–4) $\times 10^6$ data points from the detectors for each setup, we filter them using the aforementioned methods. **For the scintillators** (hereafter referred to as *the carbon environment*, as they are mostly carbon), the cuts require no adjustment besides setting the time windows and pulse height thresholds until the signal is evident. **For the water tank** (which will be called *the oxygen environment*, as it is mostly made up of oxygen), Cut 4 is no longer used as there is no third detector beyond the stopping environment. Instead, we alter Cut 5 such that S muons have a time difference between the two Cherenkov pulses $\sim 10\,\mu$s, while T muons would have more (i.e. from different muons).

3.1 Lifetime of the Muon

These filters enable us to extract the stopping muon counts vs. time in the form of a histogram, leaving us with 476 counts for carbon and 6144 counts for oxygen. The discrepancy in counts is mainly due to the increased number of events in the signal window of Cut 3 for the oxygen dataset. In both cases, we subtract the background of uncorrelated signals and use the simple exponential decay model in Equation (1) for fitting, in order to estimate the lifetime. This can also be obtained using two other methods, described below.

3.1.1 Exponential + Background Fit



(a) Decays in Carbon **(b)** Decays in Oxygen

Figure 4: The number of stopping muons as a function of the time elapsed after the A signal

Note: Even after subtraction of uncorrelated background and rejection of T muons, a residual population of early-time events remains due to trigger-correlated detector effects associated with muon stopping. Hence, events occurring within a few hundred nanoseconds of the trigger do not correspond to muon decay and do not follow an exponential time distribution: the fit is restricted to times $\gtrsim 75\,\text{ticks} = 600\,\text{ns}$.

Using the least squares method for fitting with a single exponential and finding the errors, we obtain two close results for the lifetime of the muon:

$$\tau^C_{\text{exp}} = (2.133 \pm 0.228) \times 10^{-6}\,\text{s}\,,$$

$$\tau^O_{\text{exp}} = (2.215 \pm 0.064) \times 10^{-6}\,\text{s}\,.$$

Both of these values lie within one standard deviation (1σ) away from the accepted value of $\tau_{\text{true}} = 2.197 \times 10^{-6}\,\text{s}$ [7], with the second being more precise as the dataset is significantly larger, causing the statistical uncertainty to improve, as can be seen in Figure 4.

However, not all stopping muons lead to decays. Previously, we have ignored that negative muons can also be captured in the environment [4], and we must correct for that. Considering an approximate distribution of positive to negative muons $\mu^+ : \mu^- \approx 53 : 47^{**}$ ($f_- = 0.47$), we adjust the lifetime following Equations (2) and (3). This may be applied to the carbon case, as we are provided with the known muon capture rate $\lambda^C_{\text{cap}} = 0.4 \times 10^5\,\text{s}^{-1}$, and we obtain $\frac{\tau^C_{\text{exp}}}{\tau_0} \simeq 0.96$. This corresponds to a 4% correction from our result, leading to a more appropriate lifetime estimate:

$$\boxed{\tau_\mu \approx (2.221 \pm 0.237) \times 10^{-6}\,\text{s}}\,.$$

3.1.2 Alternative Lifetime Measurements

Dividing the range: this approach involves splitting the runs from time $t = 0$ to another set time T_1, and from $t = T_1$ to a final time T. The numbers of particles are then:

$$N_1 = N_0\tau\left[1 - e^{-T_1/\tau}\right] \quad ; \quad N_2 = N_0\tau\left[e^{-T_1/\tau} - e^{-T/\tau}\right]$$

We may then identify the lifetime by the relative number of stopping muons. For simplicity, we implement this taking $T_1 = T/2$, yielding

$$\tau = \frac{T}{2\ln(N_1/N_2)}\,,$$

with an error of

$$\frac{\Delta\tau}{\tau} = \frac{T}{2(\ln(N_1/N_2))^2}\sqrt{\frac{1}{N_1} + \frac{1}{N_2}}\,.$$

Taking our total time interval and counting the number of stopping muons both sides of the midpoint, this method estimates $\tau_{\text{int}} \simeq (2.192 \pm 0.473) \times 10^{-6}\,\text{s}$, still within 1σ but with larger error.

Maximum likelihood approach: evaluates how the likelihood of τ varies for our sample, and we begin by taking the probability of a single decay occurring at time t_i, for $t = 0 \to \infty$: $P(t_i) = \frac{e^{-t_i/\tau}}{\tau}$. The likelihood is then

$$\mathcal{L} = \prod_i \frac{e^{-t_i/\tau}}{\tau} \Rightarrow \ln\mathcal{L} = \sum_i \left[-\frac{t_i}{\tau}\right] - N\ln\tau\,.$$

Maximizing this with respect to τ and with some manipulation of the result, we obtain a final form for the lifetime estimate:

$$\tau_{\mathcal{L}} = \frac{\sum t_i/N}{1 - \frac{(T/\tau_{\mathcal{L}})e^{-T/\tau_{\mathcal{L}}}}{1 - e^{-T/\tau_{\mathcal{L}}}}}$$

and a statistical error corresponding to letting $\ln\mathcal{L} \to \ln\mathcal{L}_{\max} - \frac{1}{2}$.

This represents a simplified version of the maximum likelihood method and a more complicated means of estimating the lifetime. Nevertheless, we obtain $\tau_{\text{ML}} \simeq (2.459 \pm 0.124) \times 10^{-6}\,\text{s}$, a result about 2σ away from the true value, yet with smaller uncertainty (Appendix A).

**There are more μ^+ than μ^- since cosmic rays are mostly protons, and when a proton hits the atmospheric nuclei, it produces more π^+ than π^-, as it carries positive isospin. Positive pions almost always decay like $\pi^+ \to \mu^+ + \nu_\mu$, and negative ones also have a chance of being absorbed in the nuclei before they decay.

3.2 Muon Capture Rates in Carbon and Oxygen

So far, we have ignored the capture of negatively charged muons in our stopping samples, only amending the lifetime results subsequently. That, too, involved knowing the capture rate beforehand, which was only given for carbon, but not oxygen. For that reason, this part of the experiment focuses on using an improved fitting model for our data, accounting for these *different* capture rates in the two media.

3.2.1 Initial Attempts and Limitations

We are looking to use the double exponential model described in Equation (4), combining the two datasets and fitting them simultaneously. This is reasonable, since the lifetime of the muon and the fraction of μ^- are common to both. Our initial goals are to determine the *joint* muon lifetime, the muon ratio, as well as the capture rates in carbon and oxygen. For that, we attempt three methods.

1. **Double exponential with all parameters free** — we use a coupled model, having τ_0 and f_- common free parameters, and $\lambda_{\text{cap}}^C, \lambda_{\text{cap}}^O, N_0^C, N_0^O$ also free. This produces reasonable fits apparently, however the code struggles with that many free parameters. It is impossible to properly adjust capture rates and the muon fraction as the fit progresses, causing times to increase chaotically. This essentially reduces the fits to a single exponential and causes the fraction to take extreme values, imposed by our limits.

2. **Double exponential with fixed τ_0** — the previous model is clearly not reasonable as it is unconstrained, leading us to consider a similar strategy, but now with the lifetime fixed by prior measurements. This fit is better and more physically accurate, as it does not reduce to the simple model at first, but f_- still tends toward extreme values, causing the capture rates not to be identified properly either.

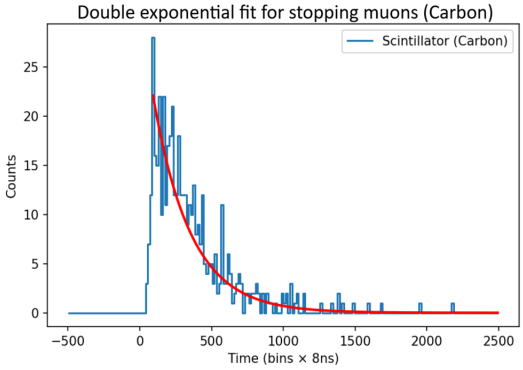
3. **Split-up exponential with fixed τ_0** — this model is more intricate, as it tries to deal with the time-changing fraction (which seems to vary at later times when mostly μ^+ remain), involving the separate treatment of the histogram's tail (Appendix B). We consider a portion at the end where we assume no μ^- remain, fitting it with a single exponential. This enables us to make an estimate for the initial f_- and, after subtracting this tail for early times, we can use this fraction to fit the remaining portion with both types of muons. Even though we could, in principle, find some actual values for the capture rates, the issue that arises is that clearly f_- will be very poorly estimated given how rough of an approximation we made. This is insufficient in the context of the minute difference between μ^+ and μ^-, and also yields too high capture times.

This leaves us to conclude that **we cannot reliably determine the fraction of negative muons** with the setup at hand. In general, this can be performed using strong magnetic fields in a magnetic spectrometer, which would make it possible to tag the signs of these particles, and may be the subject of a future investigation. The solution to the fitting issue is then to fix both τ_0 and f_-, and return to the enhanced double exponential model from before.

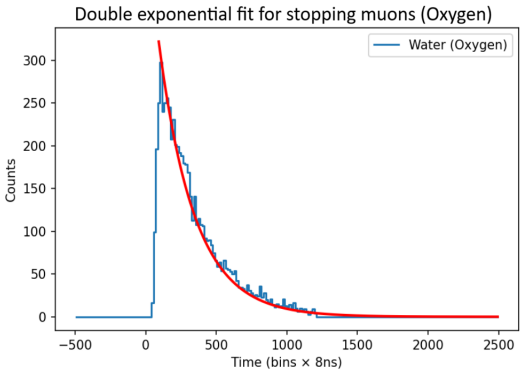
3.2.2 Capture Rate Extraction

The improved model allows us to compare the coupled fits in Figure 5[†] and also extract the different capture rates and their errors:

[†]The lifetime measurement fits from Figure 4 were made independently, hence for the more statistically accurate dataset for oxygen, the range was chosen to be narrower such as to better highlight the decay. In the case of Figure 5, the range is the same for both scenarios to properly compare them, which is why the carbon histogram looks similar to before, but the oxygen one appears different due to increased bin sizes. The data used throughout is unchanged.



(a) Decays in Carbon



(b) Decays in Oxygen

Figure 5: Enhanced fits on the number of stopping muons as a function of the time elapsed

$$\boxed{\begin{aligned}\lambda^{C}_{\text{cap}} &= (0.6 \pm 0.9) \times 10^{5}\,\text{s}^{-1}\,,\\ \lambda^{O}_{\text{cap}} &= (0.9 \pm 0.2) \times 10^{5}\,\text{s}^{-1}\,.\end{aligned}}$$

We note that the capture rate for carbon is within one standard deviation from the quoted value [8] of $\lambda^{C}_{\text{true}} = (0.44 \pm 0.10) \times 10^{5}\,\text{s}^{-1}$, yet the fit does not appropriately constrain it. This is mainly due to the capture rate being an order of magnitude smaller than the decay rate in carbon. As a result, capture behaves as a small perturbation to the dominant exponential decay, leading to strong correlations between fit parameters, thus large statistical uncertainties.

The extracted capture rate for oxygen does not suffer from this issue as it poses a larger deviation from pure decay, given the more positive nucleus. It can be confirmed from other experimental studies [8] that our result is consistent with the measured value of $\lambda^{O}_{\text{true}} = (0.98 \pm 0.03) \times 10^{5}\,\text{s}^{-1}$.

We are finally able to verify our theoretical expectation from Equation (9) that $\lambda_{\text{cap}} \propto Z^{3}$:

$$\boxed{\frac{\lambda^{O}_{\text{cap}}}{\lambda^{C}_{\text{cap}}} = 2.25 \pm 0.5 \approx 2.37 = \left(\frac{8}{6}\right)^{3} = \left(\frac{Z_O}{Z_C}\right)^{3}},$$

where we have used the measured capture rate for oxygen and the given, more accurate value for carbon $\lambda^{C}_{\text{true}}$. The error is quite significant, stemming from the uncertainty on the capture rate measurement $\Delta(\lambda^{O}_{\text{cap}}/\lambda^{C}_{\text{cap}}) = (\lambda^{O}_{\text{cap}}/\lambda^{C}_{\text{cap}})\sqrt{\left(\Delta\lambda^{O}_{\text{cap}}/\lambda^{O}_{\text{cap}}\right)^{2} + \left(\Delta\lambda^{C}_{\text{cap}}/\lambda^{C}_{\text{cap}}\right)^{2}}$, yet confirms the expected Z dependence (in reality, $\lambda_{\text{cap}} \propto Z^{n}$, with $2.5 \lesssim n \lesssim 4$, might be more accurate).

3.3 Error Analysis

Several sources of error interfere with our measurements, either systematic or statistical. Although the implemented cuts are designed to reduce noise, certain errors still get through.

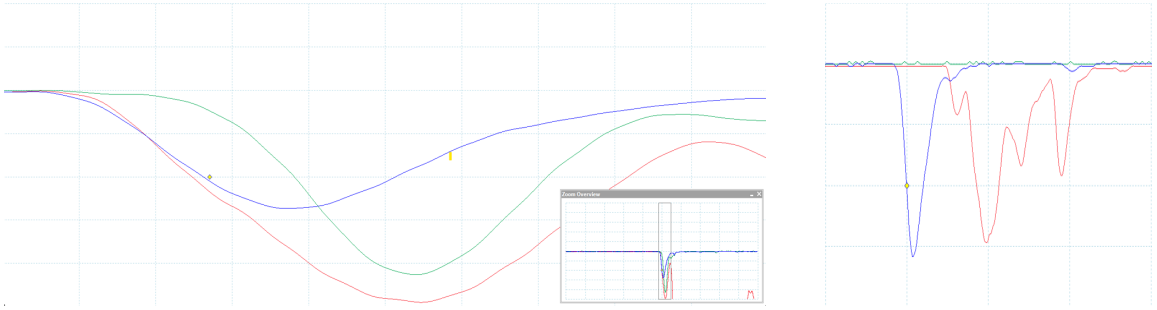
Cut 3 requires an asymmetric correlation window, which is due to a **systematic offset from cable delays or the signal propagation in the PMT**. A peak in counts is then observed on the other side of zero, coming from pulses in B that happened before A. This is also a systematic error called **pre-pulsing**, which is characterized by detector artifacts and electronics effects.

Cut 4 takes the second time window selection between signals on detectors A and C, instead of B and C, due to the position of the PMT on the cylinder detector being at the other end of the counter from the slab detectors. Otherwise, **the resolution** would be affected by a further

$d/c \approx 1\,\text{m} \,/\, 1\,(\text{ft}/\text{ns}) \approx 3\,\text{ns}$. As it stands, the resolution is given by 1 tick, being the equivalent of 8 ns — this can be confirmed by measuring the width of pulses in the Picoscope 6 software. Time differences between successive correlated peaks indicate a 10 ± 2 ns time resolution, while measurements of FWHM of the signals give 20 ± 5 ns, and we expect the resolution to be half that (Figure 6a).

Cut 6 employs a threshold to eliminate **after-pulses** — these happen as electrons produced ionize material inside the relatively old PMT, and the ions are driven back by the strong electric field and produce signals, over times $\sim \mu$s — this is similar to the muon lifetime and may be confused with decays. Such signals are visible in Figure 6b, with characteristically smaller peaks after the actual decay.

Ideally, we would not make any prior cuts on the signals from the B counter, however in our limited detector setup we are required to make the first two cuts to reduce the file size, so cut 6 ultimately leaves us with a **biased distribution**. The **height threshold** itself is quite sensitive as cutting too low leaves too many false counts, and too high leads to larger statistical errors (see Figure 3c).



(a) Sample T muon signature **(b)** Sample S muon signature

Figure 6: Signal amplitudes vs. time, as viewed in the Picoscope 6 software (Note: these correspond to the first setup, with channel A being blue, B red and C green.)

Our stopping muon fits are also affected by residual detector effects and a non-ideal selection process. We have already analyzed statistical effects such as parameter correlations, interval and threshold selection, and coincidence background, as well as physical effects like the barrier imposed by the capture rate being significantly lower than the decay rate and error propagation in our methods. We further note a number of systematic sources, namely **prompt or delayed scintillation** and **secondary energy deposition**. The latter arises due to relativistic δ-electrons ejected from atoms as muons pass through, and are different from pre- or after-pulses.

Lastly, it is relevant to analyze the **relative efficiencies of the top detectors** on which we trigger (i.e. A and D), as they define the signal rates and acceptance of our experiment. Conditional probabilities are measured by placing one on top of the other and triggering on each one separately, then checking whether one fires given that the other does. By processing samples of ~ 600 signals, implementing the first two cuts and dividing the number of triggers before and after the cuts, we highlight $\epsilon_{A|D} \simeq 99.3 \pm 1.1\,\%$ and $\epsilon_{D|A} \simeq 98.0 \pm 4.0\,\%$.

4 Conclusions and Suggestions for Further Analysis

The experiment succeeded in extracting the lifetime of the muon, as well as the capture rates of negative muons by nuclei of carbon and oxygen. Our lifetime estimate of $\tau_\mu \approx (2.221 \pm 0.237) \times 10^{-6}\,\mathrm{s}$ falls within one standard deviation of the true value of $2.197 \times 10^{-6}\,\mathrm{s}$. The capture rates we obtained from our fits for the two media, $\lambda^C_{\mathrm{cap}} = (0.6 \pm 0.9) \times 10^5\,\mathrm{s}^{-1}$ and $\lambda^O_{\mathrm{cap}} = (0.9 \pm 0.2) \times 10^5\,\mathrm{s}^{-1}$, are also consistent with accepted values of $(0.44 \pm 0.10) \times 10^5\,\mathrm{s}^{-1}$ and $(0.98 \pm 0.03) \times 10^5\,\mathrm{s}^{-1}$, respectively, and obey the theoretical model according to which $\lambda^O_{\mathrm{cap}}/\lambda^C_{\mathrm{cap}} \approx (Z_O/Z_C)^3$.

Further analysis could improve constraints on the capture rate measurements, for instance by employing goodness-of-fit checks, such as reduced χ^2 or residual plots for the fits, or by expanding the setup to a larger array of detectors to further reduce noise and unwanted effects. The present data and methods are insufficient for determining the fraction of μ^- experimentally. Such an extension to the analysis would benefit from a combination of large magnetic fields and charge-tagging spectrometers. Finally, future experiments may test capture rates in different media or with alternating layers of material, measure detector response, muon time-of-flight distribution, or even the angular spread of the muon flux, all of which would yield valuable insights into the understanding of cosmic muons, as other works have also proven [9] [10].

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A Maximum Likelihood Method

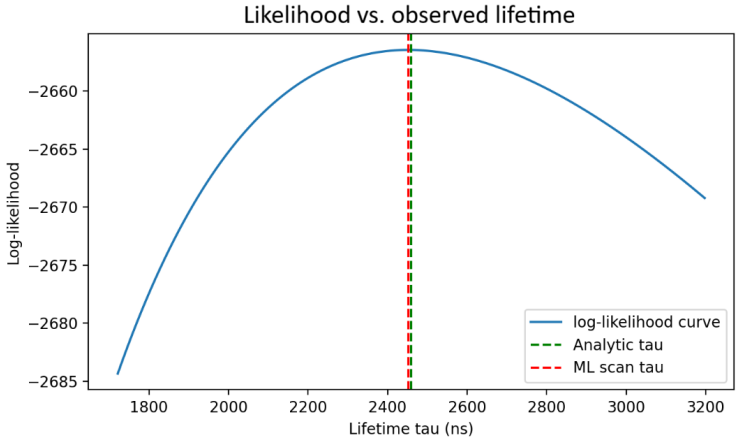


Figure 7: An alternative approach to the lifetime estimate, involving a simplified version of identifying the maximum likelihood (scintillator data)

B Split-up Exponential Approach

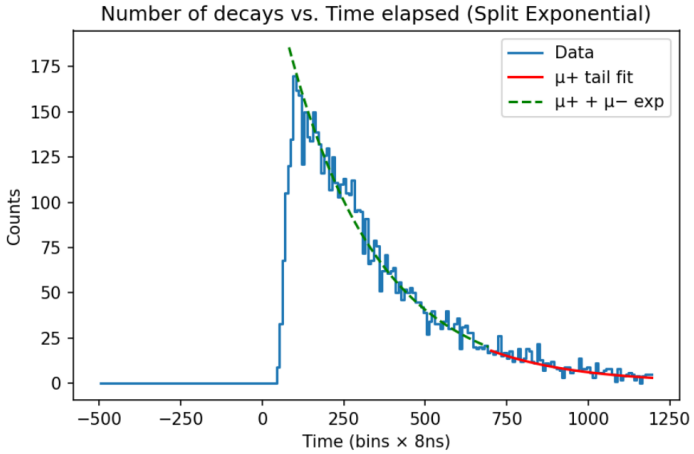


Figure 8: Enhanced fitting method for the stopping muon counts vs. time (water tank data)

C Code used in the Analysis

The relevant code fragments used throughout this project can be accessed via github, including code for **Initial Lifetime Results in Carbon**, **Lifetime in Oxygen**, and **Measuring Capture Rates**.